

Measurement Map

Companion to the manuscript skeleton, 10 October 2026

1 Scope

The map takes each object of the model in `MANUSCRIPT.tex` to its counterpart in the Basel capital framework [Basel Committee on Banking Supervision, 2011], the Basel liquidity standard [Basel Committee on Banking Supervision, 2013] and the Call Report instructions for regulatory capital [Federal Financial Institutions Examination Council, 2026], and says how it is observed. The regulatory cap is that of Bayraktar et al. [2026] (BCVW). Quotation and reading IDs refer to the `lit/` records, where each quotation is checked verbatim on its page. The rearrangements of the regulatory ratios are proved in `lean/mathlib/MeasurementMap.lean`.

A status of Rule means a regulation sets the value. Report means a regulatory report records the object for each bank. Calibration means the model takes a value with no regulatory counterpart. Unobserved means neither a rule nor a report gives it. Output means the model produces it and its observation is part of the empirical design.

2 Map

Table 1: Model objects and their observation.

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G1	x	The state, assets over liabilities.	Total assets over total liabilities. From the reported leverage ratio ℓ (Tier 1 over total assets for the leverage ratio), $x = 1/(1 - \ell)$ holds only if Tier 1 equals book equity and the denominator equals total assets.	Report	RCR-Q4, RCR-D3, <code>MeasurementMap.leverage_of_state</code>
G2	π	The share of assets in the risky asset.	Risk-weighted assets over total assets, under risk weights of 0 on the riskless and 1 on the risky asset. With a positive weight on the riskless asset the identification fails.	Report	RCR-Q5, RCR-D2, <code>MeasurementMap.rwa_two_assets</code> , <code>MeasurementMap.control_rwa_needs_zero_weight_on_riskless</code>
G3	$X - L$	Equity, assets less liabilities.	CET1 capital or Tier 1 capital as reported. Both are net of regulatory adjustments, so they equal book equity only when the adjustments vanish.	Report	B3-Q3, RCR-Q1, RCR-Q3, B3-D2
G4	a_1	The floor on equity over risky assets, the solvency branch $(x - 1)/a_1$ of the cap.	The CET1 minimum of 4.5% of risk-weighted assets, which is BCVW's baseline 0.045, or the Tier 1 minimum of 6.0%. The bank's own ratio is reported as CET1 or Tier 1 capital over risk-weighted assets. Banks electing the community bank leverage ratio report neither.	Rule	BCVW-Q9, BCVW-Q13, B3-Q1, B3-Q2, B3-D1, RCR-Q2, RCR-Q6, RCR-Q7, RCR-D1, <code>MeasurementMap.solvency_iff</code> , <code>MeasurementMap.control_solvency_needs_positive_exposure</code>

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G5	a_2	Net 30-day outflows as a share of liabilities, in the liquidity branch $(x - a_2)/a_3$ of the cap.	A run-off rate on all liabilities. BCVW's baseline 0.05 is the factor for stable insured retail deposits, and less stable retail deposits run off at 10% or more. The LCR must be at least 100%. No report read gives the rate.	Rule	BCVW-Q10, LCR-Q1, LCR-Q9, LCR-Q10, LCR-D1, MeasurementMap.lcr_iff, MeasurementMap.control_lcr_needs_positive_runoff
G6	a_3	The haircut on the risky asset in high-quality liquid assets, the slope $1/a_3$ of the liquidity branch.	The riskless asset counts in full, as Level 1 assets do. The standard's haircuts are 15% (Level 2A) and 25% or 50% (Level 2B). BCVW's baseline 0.30 is none of them. The 40% cap on Level 2 assets and the 75% cap on inflows are not in the model.	Calibration	BCVW-Q11, BCVW-Q12, BCVW-Q13, LCR-Q2, LCR-Q3, LCR-Q4, LCR-Q5, LCR-Q6, LCR-Q7, LCR-Q8, LCR-D2, LCR-D3, MeasurementMap.hqla_haircut, MeasurementMap.cap_iff
G7	q	The requirement on $x - 1$, with $z = 0$ at $x = 1 + q$.	A floor ℓ_0 on the leverage ratio is the floor $\ell_0/(1 - \ell_0)$ on $x - 1$, so the 3% Tier 1 leverage ratio gives $q = 3/97$ when Tier 1 equals book equity. This is a reading, and the bundle does not fix q . The z -volatility does not depend on q .	Rule	B3-Q7, RCR-Q4, MeasurementMap.leverage_req_iff, MeasurementMap.control_leverage_needs_10_lt_one, MeasurementMap.zvol2_free_of_q
G8	σ, μ_s, r	The volatility and drift of the risky asset and the riskless rate.	No regulatory counterpart. BCVW's baseline is $\sigma = 0.08, \mu = 0.04, r = 0.02$.	Calibration	BCVW-Q14
G9	$\sigma_L, c, \gamma, \mu_L, r_L$	The liability shock, its correlation with the risky return, the net income on liabilities, the liability drift and the deposit rate, with $\mu_L = \gamma + r_L$.	No regulatory counterpart. BCVW's baseline is $\sigma_L = 0.03, c = 0.20, \gamma = 0.02, \mu_L = 0.03$.	Calibration	BCVW-Q14, M9
G10	ρ_L	The liability-net discount rate of the reduced problem.	Formed from the shareholder discount rate, BCVW's baseline $\rho = 0.12$. No regulatory counterpart.	Calibration	BCVW-Q14
G11	κ	The proportional issuance cost.	No regulatory counterpart. BCVW's baseline is $\kappa = 0.01$ for dilution and $\kappa' = 0.02$ per unit issued.	Calibration	BCVW-Q14
G12	K	The fixed issuance cost, charged per unit of liabilities.	Not observed. A cost fixed in dollars would break the reduction to x . Locally identified together with λ_S from the barrier and the trigger.	Unobserved	BCVW-Q4, K2, M5, M6
G13	λ_S	The asymmetry parameter of the shortfall penalty.	Not observed. The variance ratio carries no information about it. Locally identified together with K from the barrier and the trigger.	Unobserved	K1, K2, M3, M6, M11
G14	y^*	The dividend barrier.	The state at which the bank pays dividends. Under the conservation buffer a bank with CET1 between 4.5% and 7.0% of risk-weighted assets has its distributions restricted, so observed payouts reflect the rule as well as the barrier.	Output	M4, B3-Q4, B3-Q5, B3-Q6, B3-D3
G15	x_L, y_{post}	The recapitalisation trigger and the injection target.	The state just before and just after an equity issue.	Output	M5, M6, M7, M10

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G16	ζ, v_1^2, v_3^2	The z -volatility and its limits at the distress boundary and deep in surplus.	The variance of changes in $z = \log((x-1)/q)$ near the distress boundary and deep in surplus. It does not depend on q .	Output	M2, M3, M8, MeasurementMap.zvol2_free_of_q
G17	v_1^2/v_3^2	The deficit-to-surplus variance ratio, read in discrete time as λ_V^4 .	The ratio of the two variances of G16. Reading it as λ_V^4 is a definition.	Output	K1, M1, M3

3 The ratio constraints in the state

With equity $X - L$, risk-weighted assets πX , high-quality liquid assets $X - a_3 \pi X$ and net outflows $a_2 L$, the two ratio constraints of BCVW are the two branches of the cap in $x = X/L$. For $a_1, a_2, a_3, \pi, X, L > 0$,

$$a_1 \leq \frac{X-L}{\pi X} \iff \pi x \leq \frac{x-1}{a_1}, \quad 1 \leq \frac{X-a_3 \pi X}{a_2 L} \iff \pi x \leq \frac{x-a_2}{a_3},$$

and both hold if and only if $\pi x \leq u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ (MeasurementMap.solvency_iff, MeasurementMap.lcr_iff, MeasurementMap.cap_iff). At $\pi = 0$ the left side of the first equivalence is undefined and the equivalence fails, and at $a_2 = 0$ the same holds for the second (the two controls). For $\ell_0 < 1$ and $X, L > 0$, $\ell_0 \leq (X-L)/X \iff \ell_0/(1-\ell_0) \leq x-1$ (MeasurementMap.leverage_req_iff).

4 Pending

- The FDIC field names used by the empirical scripts (RWAJT, RBCT1J, RBCRWJ, ASSET, LIAB, EQCDIV, EQCSTKRX) do not appear in the Call Report instructions read. Their link to the items of G1 to G4, G14 and G15 needs the FDIC data dictionary.
- Rows G1 to G3 assume regulatory capital equals book equity. Items 12 to 18 and 27 to 29 of Schedule RC-R, which give the adjustments, were not read.
- The tier that a_1 stands for is not fixed (B3-D1).

References

- Basel Committee on Banking Supervision. Basel III: A global regulatory framework for more resilient banks and banking systems. Technical report, Bank for International Settlements, June 2011. December 2010, revised June 2011.
- Basel Committee on Banking Supervision. Basel III: The liquidity coverage ratio and liquidity risk monitoring tools. Technical report, Bank for International Settlements, January 2013.
- Erhan Bayraktar, Etienne Chevalier, Vathana Ly Vath, and Yuqiong Wang. Tractable bank capital structure: Optimal control under Basel III constraints, 2026. arXiv:2603.14557v2.
- Federal Financial Institutions Examination Council. *Instructions for Preparation of Consolidated Reports of Condition and Income (FFIEC 031 and 041), Schedule RC-R, Part I: Regulatory Capital Components and Ratios*. Federal Financial Institutions Examination Council, June 2026.